

Yuan Yin

 yuan-yin-nn  yuan-yin.github.io  French, English, Mandarin  Paris, France



Professional Profil

Passionate about cutting-edge AI technologies, especially in **Machine Learning** (ML) and **Deep Learning** (DL), I specialize in pioneering **neural network** methods for analyzing physical dynamics. This expertise allows me to develop DL/AI solutions for complex real-world challenges by innovating upon existing methods, and integrating DL approaches into existing non-ML systems.

Experience

- Valeo.ai** *Paris, France*
- ▶ **AI Researcher** Dec 2024 ▶ Now
 - ▶ **Postdoctoral AI Researcher** Apr 2024 ▶ Nov 2024
Corner-case generation for learning robust self-driving cars
- Sorbonne Université, ISIR, MLIA Team** *Paris, France*
- ▶ **Postdoctoral Researcher** Jul 2023 ▶ Dec 2023
Supervising ongoing research projects & making a tutorial on physics-aware DL
 - ▶ **PhD Student, Teaching Assistant** Oct 2019 ▶ Jun 2023
Supervised by Patrick Gallinari & Nicolas Baskiotis
Physics-Aware DL and dynamical systems: DL-physics hybrid modeling; Out-of-distribution generalization for dynamics modeling; Continuous dynamics modeling
 - ▶ **Research Intern in Deep Learning** Feb 2019 ▶ Sep 2019
Imputing spatiotemporal data with generative models
- Inria Paris Research Intern in NLP** Feb 2018 ▶ Jul 2018
- Beihang Univ. Research Intern in CV** May 2015 ▶ Jun 2016

Education

- Sorbonne Université** *fka UPMC (Paris-6)* *Paris, France*
- ▶ **PhD in Machine Learning and Deep Learning** Jun 2023
 - ▶ **MSc2 DAC Master Data Science Paris** 2019
- Université Paris Cité** *fka Paris-Diderot (Paris-7)* *Paris, France*
- ▶ **MSc1 MPRI Parisian Research Master in Comp. Sci.** 2018
 - ▶ **Univ. Dipl. in French Language and Civilization** 2017
- Beihang University** *Beijing, China*
- ▶ **BSc in Computer Science** 2016

Technical Skills

- | | |
|--------------------------|---|
| OS & Platform | Linux servers equipped with GPUs |
| Programming | Python (PyTorch, JAX, etc.), C/C++, Java, $\text{\LaTeX}$, Matlab, OCaml |
| Tools | Git, Emacs, VS Code, Eclipse |

Languages

- | | | | |
|-----------------|-------------------|-----------------------|-----------|
| French | Bilingual | last exam ▶ C1 (2017) | ● ● ● ● ● |
| English | Full Professional | last exam ▶ B2 (2015) | ● ● ● ● ● |
| Mandarin | Native | | ● ● ● ● ● |

Distinctions

Accessit for the 2024 AI Thesis Prize from the **French Association for Artificial Intelligence (AFIA)**

Top Reviewer at NeurIPS 2023

Community Service

Conference Reviewer at NeurIPS 2021–25, ICLR 2023–26, ICML 2022–25, CVPR 2025, ICRA 2026, ECML-PKDD 2021, and ACM Multimedia 2021

Workshop Reviewer at CCFM at NeurIPS 2025, ML4PS at NeurIPS 2022–24, Physics4ML at ICLR 2023, SynS&ML at ICML 2023, and ROAM at ECCV 2024

Teaching in French at Sorbonne Université (2019–22) in Engineering Department (UFR 919). For undergrads: C Programming (L1), Algorithmics (L2), Probabilities (L3). For postgrads: ML Research Methodology (M2)

Publications

Conference and Journal Papers

** Equal contribution*

- **Y. Yin**, S. Venkataramanan, T.-H. Vu, A. Bursuc, and M. Cord. IPA: An information-reconstructive input projection framework for efficient foundation model adaptation. In *TMLR*.
(Oral and Best Paper at NeurIPS 2025 CCFM WS)
- L. Le Boudec, E. de Bézenac, L. Serrano, R. D. Regueiro-Espino, **Y. Yin**, and P. Gallinari. Learning a neural solver for parametric PDE to enhance physics-informed methods. In **ICLR 2026**.
- A. Kassā Koupaï, J. Mifsut-Benet, **Y. Yin**, J.-N. Vittaut, and P. Gallinari. Boosting generalization in parametric PDE neural solvers through adaptive conditioning. In **NeurIPS 2024**.
- E. Le Naour, L. Serrano, L. Migus, **Y. Yin**, G. Agoua, N. Baskiotis, P. Gallinari, and V. Guigue. Time series continuous modeling for imputation and forecasting with implicit neural representations. *TMLR*, 2024.
- **Y. Yin***, M. Kirchmeyer*, J.-Y. Franceschi*, A. Rakotomamonjy, and P. Gallinari. Continuous PDE dynamics forecasting with implicit neural representations. In **ICLR 2023**.
(Spotlight)
- L. Serrano, L. Le Boudec, A. Kassā Koupaï, **Y. Yin**, T. X. Wang, J.-N. Vittaut, and P. Gallinari. Operator learning with neural fields: Tackling PDEs on general geometries. In **NeurIPS 2023**.
- C. Metta, A. Beretta, R. Guidotti, **Y. Yin**, P. Gallinari, S. Rinzivillo, and F. Giannotti. Improving trust and confidence in medical skin lesion diagnosis through explainable deep learning. *Int. J. Data. Sci. Anal.*, 2023.
- M. Kirchmeyer*, **Y. Yin***, J. Donà, N. Baskiotis, A. Rakotomamonjy, and P. Gallinari. Generalizing to new physical systems via context-informed dynamics model. In **ICML 2022**.
(Spotlight)
- **Y. Yin**, I. Ayed, E. de Bézenac, N. Baskiotis, and P. Gallinari. LEADS: Learning dynamical systems that generalize across environments. In **NeurIPS 2021**.

· **Y. Yin***, V. Le Guen*, J. Donà*, E. de Bézenac*, I. Ayed*, N. Thome, and P. Gallinari. Augmenting physical models with deep networks for complex dynamics forecasting. In **ICLR 2021**.
(Oral, also in J. Stat. Mech.: Theory Exp.)

· D. Huang, R.K. Zhang, **Y. Yin**, Y.D. Wang, and Y.H. Wang. Local feature approach to dorsal hand vein recognition by centroid-based circular key-point grid and fine-grained matching. *Image Vis. Comput.*, 2017.

Workshop Papers

· Y.H. Xu*, **Y. Yin***, T.-H. Vu, A. Boulch, É. Zablocki, and M. Cord. PPT: Pre-training with pseudo-labeled trajectories for motion forecasting. In *CoRL 2025 Workshop on Robot Data*, 2024.

· **Y. Yin**, P. Khayatan, É. Zablocki, A. Boulch, and M. Cord. ReGentS: Real-world safety-critical driving scenario generation made stable. In **ECCV 2024** *Workshop on W-CODA*.

· L. Serrano, L. Migus, **Y. Yin**, J. A. Mazari, J.-N. Vittaut, and P. Gallinari. IN-FINITY: Neural field modeling for reynolds-averaged navier-stokes equations. In **ICML 2023** *Workshop on SynS & ML*.

· L. Migus, **Y. Yin**, J. A. Mazari, and P. Gallinari. Multi-scale physical representations for approximating PDE solutions with graph neural operators. In **ICLR 2022** *Workshop on GTRL*.

· **Y. Yin**, A. Pajot, E. De Bézenac, and P. Gallinari. Unsupervised inpainting for occluded sea surface temperature sequences. In *CI 2019*.

Preprints

· **Y. Yin**, A. Pajot, E. de Bézenac, and P. Gallinari. Unsupervised spatiotem-poral data inpainting, 2020.

 Presentations and Invited Talks

Please find the details of the talks on [my website](#)

Oral Workshop Presentation at NeurIPS 2025 CCFM Workshop	dec. 2025
In-Person Poster Session at ECCV 2024	Sep 2024
Workshop on <i>Mathematical Foundations of AI</i> at DATAIA-SCAI	Jan 2024
Seminar at Valeo.ai	Jan 2024
Seminar UMR MIA Paris-Saclay at AgroParisTech	Nov 2023
Seminar LAGA-MCS at Univ. Sorbonne Paris Nord	Nov 2023
Tutorial at ECML-PKDD 2023	Sep 2023
PhD Defense	Jun 2023
Seminar of Signal Processing Lab (LTS4) at EPFL	May 2023
Spotlight Conference Presentation at ICLR 2023	May 2023
AI4Science Talks at ML for Simulation Lab at Univ. of Stuttgart & NEC Labs Europe	Apr 2023
SIG LearnFluidS at d’Alembert, Sorbonne Univ.	Mar 2023
Medical Biology Engineers Day of AP-HP	Mar 2023
Seminar at Criteo AI Lab	Nov 2022
Seminar Sorbonne-ISAE-CERFACS	Oct 2022
Spotlight Conference Presentation at ICML 2022	Jul 2022
Seminar at Extrality (Now Ansys SimAI)	Feb 2022
Conference Presentation at NeurIPS 2021@Paris	Dec 2021
AAAI 2021 Spring Symposium MLPS	Mar 2021